

SMART INDIA
HACKATHON
2026

SMART INDIA HACKATHON 2026 · PROBLEM STATEMENT SIH26090

PROBLEM STATEMENT TITLE

AI-Driven Market Linkage and Smart Cataloging Mobile Application for Marginalized Artisans

CraftLink AI a virtual business manager in the artisan's pocket

One photo. One spoken sentence. A market-ready listing.

PROBLEM STATEMENT ID

SIH26090

PS CATEGORY

Software

ORGANISATION

Ministry of Social Justice and Empowerment (MoSJE)

THEME

Miscellaneous

TEAM ID

126401u

TEAM NAME

Team Zero Click



IDEA TITLE



THE GAP TODAY

7 M

artisans in India — the vast majority still outside digital commerce.

60%

of product value captured by middlemen when they handle the online sale.

3 wks

of market access a year, through melas like Dilli Haat and Surajkund.



THE SOLUTION

CraftLink AI turns a product photo and one spoken sentence into a market-ready, multilingual, fairly-priced listing.



Photo capture

Background removal, lighting correction, e-commerce framing.



Voice, in any local language

Whisper speech-to-text, running offline — no typing, no English.



AI cataloging

SEO title, description, category and keywords.



Explained price

Materials, labour, market bands and signals read from the photo and description.



Market linkage

ONDC/GeM-ready fields; bulk buyers enquire and the artisan quotes back in-app.

Human-in-the-loop: nothing publishes until the artisan approves the words and the price.



HOW IT ADDRESSES THE PROBLEM

No professional photos

→ AI removes clutter, fixes lighting and formats to e-commerce standards.

No English, low digital literacy

→ Local-language voice notes become SEO-ready descriptions.

No pricing knowledge

→ ML proposes a price from materials, labour and market trends — with reasons.

Only seasonal melas

→ A 365-day storefront replaces a few exhibition weeks a year.

Middlemen take the margin

→ Direct artisan-to-buyer linkage through ONDC and Bharat-TULIP.

UNIQUE VALUE PROPOSITION

Not another marketplace — an AI co-seller that does the photography, copywriting and pricing an artisan cannot, then hands the finished listing to the rails that already exist: ONDC and MoSJE's Bharat-TULIP.

TECHNICAL APPROACH



TECHNOLOGY STACK

AI & ML

Computer vision — rembg / U²-Net, OpenCV
Speech-to-text — faster-whisper (offline)
NLLB-200 — offline translation, 9 languages
Scikit-learn — price band model

FRONTEND

React + Vite + Tailwind
Installable PWA + Android APK (Capacitor)

BACKEND & DATA

Python + FastAPI
SQLite / PostgreSQL + SQLAlchemy

INTEGRATIONS

ONDC/GeM-ready catalogue fields
In-app B2B bulk-quote workflow
REST + JSON · .env secrets

Build status: photo → voice → 9-language listing → price → order → bulk quote, offline, today.

1

INPUT

Product photo
Voice note in the
local language

2

AI UNDERSTANDING

Image clean-up
Speech-to-text
Translation

3

PRODUCT INTELLIGENCE

Attribute extraction
Listing copy, 9 languages
Category & keywords

4

MARKET INTELLIGENCE

Explained price
Demand signals
Buyer matching

5

MOBILE + WEB OUTPUT

Artisan app + dashboard
ONDC / Bharat-TULIP
publish



FASTAPI / REST API request handling · business logic · orchestration · auth



ARTISAN DATA
profiles, skills, clusters



PRODUCT DATA
images, details, categories



MARKET DATA
trends, prices, demand



BUYER DATA
preferences, history



HOW IT DELIVERS VALUE



Artisan speaks



AI builds listing



Priced & published



Buyer matched



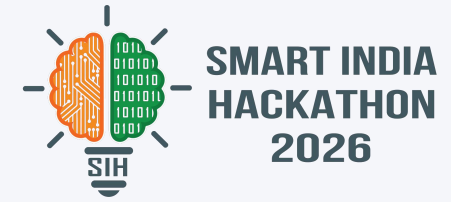
Income rises



CORE INNOVATION

One pipeline, craft to commerce. A photo and a spoken sentence become a priced, multilingual, ONDC-ready listing — with the artisan in control at every step.

FEASIBILITY AND VIABILITY



TECHNICAL FEASIBILITY

AI integration Vision, Whisper ASR, NLLB-200 and ML automate clean-up, copy and pricing.

Mobile-first delivery React + FastAPI, shipped as an Android APK and an offline-capable PWA.

Multilingual by default Voice input removes typing and digital-literacy barriers.

Proven stack Every component is open-source and already running in our build.



OPERATIONAL FEASIBILITY

Four taps to sell Open → photograph → speak → approve. No forms, no English.

Works on a shared phone Entry-level Android; listings queue and sync when signal returns.

Cluster-led rollout Onboard through existing artisan clusters and MoSJE corporations.

Artisan stays in control Every AI output is reviewed and editable before publishing.



ECONOMIC VIABILITY

Zero cost to the artisan No listing fee, no commission — the margin stays with the maker.

Low-cost to operate Every model runs offline on our own server; no per-call API bill.

Publicly fundable Fits NSFDC / NBCFDC / NSKFDC and CSR funding routes.

Scales without rebuild New languages, crafts and markets are configuration, not code.



CHALLENGES

Image quality Dim, cluttered phone photos reduce segmentation accuracy.

Voice & dialect Accent and dialect variation degrades transcription.

AI reliability Generated copy can misdescribe a craft or its provenance.

Thin market data Sparse comparables weaken any pricing signal.



MITIGATION STRATEGIES

Guided capture On-screen framing hints plus automatic enhancement before inference.

Two-pass ASR A fast Whisper pass, with a large-v3 re-run whenever confidence drops.

Artisan approval gate Nothing publishes unreviewed; corrections retrain the prompts.

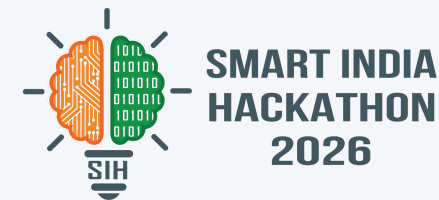
Evidence-labelled pricing Show the basis for every number; never invent a comparable.



DELIVERY ROADMAP

- **NOW**
Photo → voice → 9-language listing → price → order → bulk quote, end to end.
- **0-3 MONTHS**
Cluster pilot on the Android APK — 9 languages, offline.
- **3-9 MONTHS**
Bhashini APIs for the remaining scheduled languages; certified ONDC onboarding.
- **9-18 MONTHS**
Scale via NSFDC, NBCFDC and NSKFDC networks.

IMPACT AND BENEFITS



7 M

artisans addressable across India

60%

of value today lost to middlemen

9

Indian languages in the app today

0%

commission taken from the artisan



IMPACT ON STAKEHOLDERS

- **MoSJE beneficiary artisans** SC, OBC, DNT, Safai Karamcharis and PwD makers
Year-round digital market access without English, a camera or digital skills.
- **Buyers and customers** retail and institutional
Authentic, discoverable crafts with GI-aware categories and pricing they can see the basis for.
- **Artisan communities and clusters** across 455 craft categories
Traditional skills gain visibility and commercial value without leaving the village.
- **MoSJE and its corporations** NSFDC · NBCFDC · NSKFDC
A measurable, always-on digital channel that extends Bharat-TULIP beyond seasonal melas.



BENEFITS OF OUR SOLUTION

- **Higher market reach** A 365-day storefront replaces a few exhibition weeks a year.
- **Better earning potential** Direct linkage returns the margin that middlemen take today.
- **Language-inclusive commerce** Voice, screen and listings in 9 Indian languages, offline.
- **Faster onboarding** Photo and voice replace a day of English form-filling.
- **Smarter matching** Categories, GI tags and demand trends drive buyer discovery.



REAL-WORLD OUTCOME



More visibility



Higher income



Business growth



Stronger clusters



Skills preserved



GOVERNMENT SOURCES & SCHEMES

1

Ministry of Social Justice and Empowerment, Government of India.

Bharat-TULIP (Traditional Artisans' Upliftment Livelihood Programme) — e-marketplace for marginalized artisans; launched 5 Nov 2024, 1,500+ artisans onboarded.

<https://nbcfdc.gov.in/nbcfdc/web/tulip>

2

Ministry of Social Justice and Empowerment, Government of India.

PM-DAKSH — skilling and livelihood support for SC, OBC, EWS, DNT and Safai Karamchari artisans.

<https://socialjustice.gov.in/>

3

Development Commissioner (Handicrafts), Ministry of Textiles.

Crafts Clusters of India — artisan development, marketing and cluster support; India Handmade marketplace.

<https://handicrafts.nic.in/>

4

India Brand Equity Foundation, Ministry of Commerce & Industry.

Indian handicrafts industry — ~7 million artisans; exports ₹33,122 crore in FY 2024-25.

<https://www.ibef.org/exports/handicrafts-industry-india>



STANDARDS, PLATFORMS & DATA

5

Bhashini — National Language Translation Mission, MeitY.

ASR, machine translation and TTS across the 22 scheduled Indian languages; 300+ models.

<https://bhashini.gov.in/>

6

Open Network for Digital Commerce (ONDC), DPIIT-backed initiative.

Seller Network Participant onboarding, digital cataloguing and marketplace discovery.

<https://www.ondc.org/>

7

Intellectual Property India, Government of India.

Geographical Indications — protection of handicrafts and traditional products; 318 GI-tagged crafts.

<https://ipindia.gov.in/>

8

Smart India Hackathon 2026, AICTE / Ministry of Education.

Problem Statement SIH26090 — AI-driven market linkage and smart cataloging for marginalized artisans.

<https://www.sih.gov.in/>

DESIGNED TO PLUG INTO INDIA'S DIGITAL PUBLIC INFRASTRUCTURE



Bhashini
on the roadmap



ONDC
market reach



Bharat-TULIP
MoSJE marketplace



GI Registry
craft authenticity



PM-DAKSH
artisan skilling